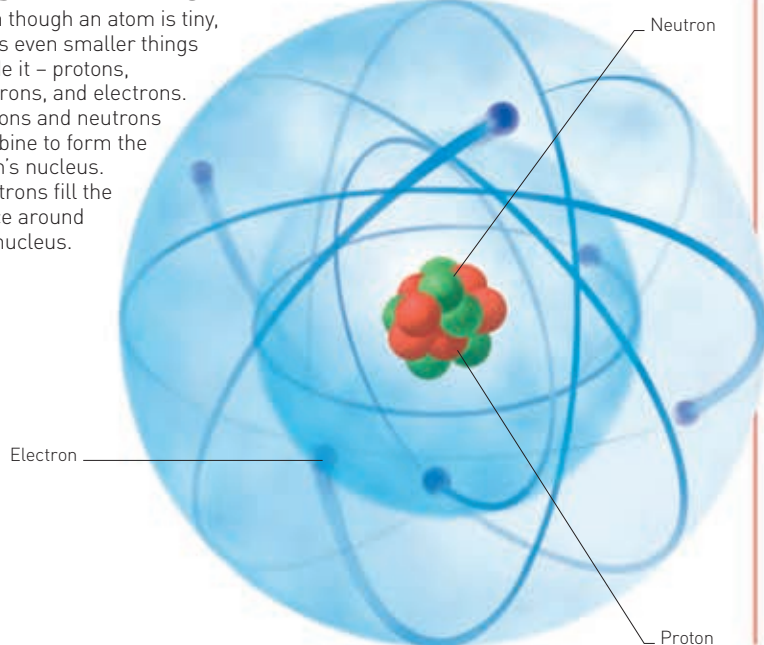


Chemistry

Chemists dig deep. They begin with the elements that make up all matter, and break them down into tiny atoms. They analyse what the atoms are, how they change state, and how they react when they mix.

INSIDE AN ATOM

Even though an atom is tiny, it has even smaller things inside it – protons, neutrons, and electrons. Protons and neutrons combine to form the atom's nucleus. Electrons fill the space around the nucleus.



CHEMISTRY IN ACTION

In ancient times, people used the natural materials around them, such as wood and stone, to make objects. Since then, scientists have discovered thousands of chemicals, some of which can be used to make new materials.



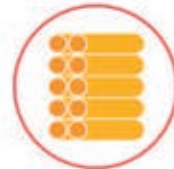
BIOCHEMISTRY
This looks at chemical processes inside living things or affecting them.



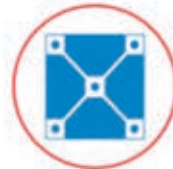
ORGANIC CHEMISTRY
This branch of chemistry focuses on carbon-based compounds and their uses.



HOUSEHOLD CHEMICALS
We use lots of chemicals in our homes, from the paint on our walls to the shampoo for our hair.



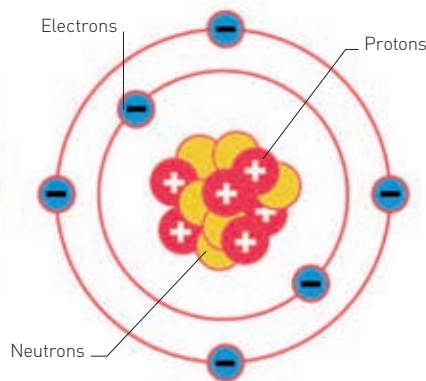
MATERIALS SCIENCE
This science uses physics and chemistry to create new materials.



ENGINEERING
Engineers use their knowledge of materials to design things.

STRUCTURE OF AN ATOM

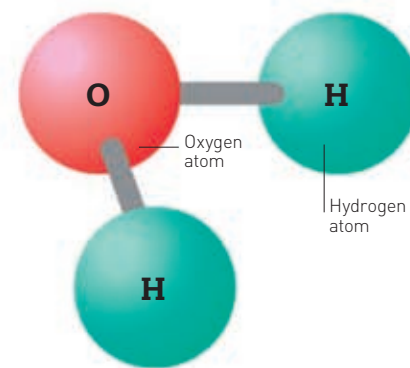
Some particles in the atom are electrically charged. The protons in the nucleus are positively charged and the orbiting electrons are negatively charged. There are always equal numbers of protons and electrons.



CARBON ATOM
The number of protons inside an atom determines what kind of atom it is. For example, a carbon atom has six electrons and six protons.

MOLECULES

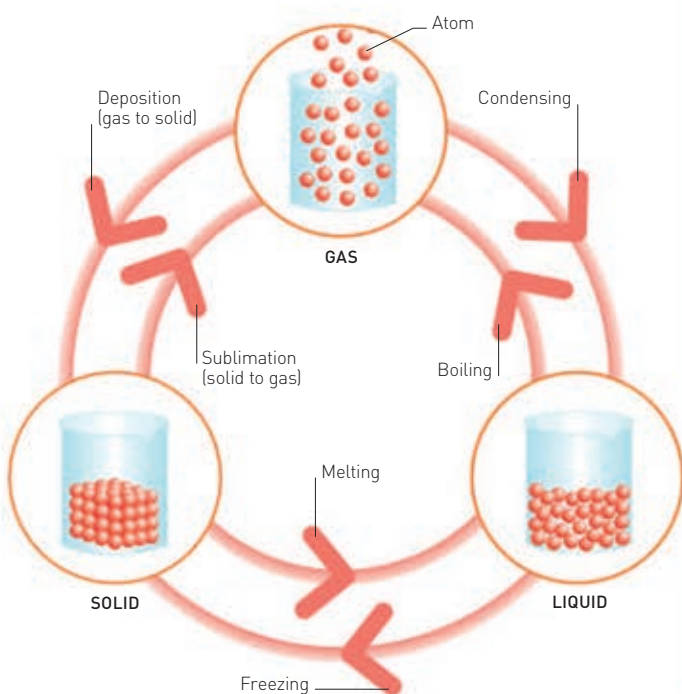
Atoms of the same sort or different atoms can clump together to make molecules. A molecule can be as simple as just two atoms, as in hydrogen, or lines of thousands of atoms, as in some plastics.



WATER MOLECULE
A molecule of water is made up of two different kinds of atoms: two hydrogen (H) atoms and one oxygen (O) atom.

STATES OF MATTER

All matter can change state. Water, for instance, can be a liquid, gas (steam), or solid (ice). Its state depends upon the way its atoms move around. As a solid, its atoms lock tightly together. As a liquid, they move further apart, and as a gas they move freely and independently.



MIXTURES

A mixture is made when two substances are combined, but no chemical reaction takes place. The ingredients are said to combine, rather than to bond.



SOLUTION
Fruit concentrate (solute) dissolves in water (the solvent) to make a drink.



SUSPENSION
A mixture between a liquid and particles of a solid, such as water and soil.



COARSE MIXTURE
An unevenly distributed mixture of different types of larger particles.



ALLOY
A mixture of a metal with other elements that creates a stronger material.

SEPARATING MIXTURES

The substances in a mixture are not bonded together, so they can be separated. However, the more similar the properties of each substance are to one another, the harder it is to separate them.



FLOATING
Shaken together these substances mix. Left for a time, they separate back out.



MAGNETIZING
Magnetic substances will be drawn to stick to the magnet.



CHROMATOGRAPHY
Using a substance that attracts some particles more than others separates the two.



FILTERING
Solid particles will collect on the filter during the filtration process.